

Problems on Median and Quartiles

1)

According to the census of 1981, following are the population figures in thousand, of 10 cities :
2000, 1180, 1785, 1500, 560, 782, 1200, 385, 1123, 222.

Find the median.

2)

Find the median from the following table :

$x :$	5	7	9	11	13	15	17	19
$f :$	1	2	7	9	11	8	5	4

3)

Calculate the mean and median from the following table :

<i>Class interval</i>	<i>Frequency</i>
6.5—7.5	5
7.5—8.5	12
8.5—9.5	25
9.5—10.5	48
10.5—11.5	32
11.5—12.5	6
12.5—13.5	1

4)

Compute the median from the following data :

<i>Mid-value</i>	<i>Frequency</i>	<i>Mid-value</i>	<i>Frequency</i>
115	6	165	60
125	25	175	38
135	48	185	22
145	72	195	3
155	116		

5)

Find the median, quartiles, 7th decile and 85th percentile from the following data :

<i>Monthly Rent</i> (₹)	<i>No. of families</i>	<i>Monthly Rent</i> (₹)	<i>No. of families</i>
20—40	6	120—140	15
40—60	9	140—160	10
60—80	11	160—180	8
80—100	14	180—200	7
100—120	20		

6)

An incomplete frequency distribution is given as follows :

<i>Variable</i>	<i>Frequency</i>	<i>Variable</i>	<i>Frequency</i>
10—20	12	50—60	?
20—30	30	60—70	25
30—40	?	70—80	18
40—50	65	Total	229

Given that the median value is 46, determine the missing frequencies using the median formula.

7)

Find the median, lower and upper quartiles, 4th decile and 60th percentile for the following distribution :

<i>Marks</i>	<i>No. of students</i>	<i>Marks</i>	<i>No. of students</i>
0—4	10	14—18	5
4—8	12	18—20	8
8—12	18	20—25	4
12—14	7	25 and above	6

[**Hint.** Here the class-intervals are not all equal. To find any partition value, there is no need to make them equal.]

Ans:

- 1) 1151.5
- 2) 13
- 3) Mean=9.87, Median=9.97
- 4) 153.8
- 5) 110,78,18,140,133.33,160
- 6) 34,45
- 7) 10.89,6.5,18.125,9.33,12.57